

AFL: Democratizing access to Australian football skills with Google Cloud

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ABOUT AFL

The AFL and Google Cloud delivered a machine learning-powered Skills Lab that enables users to access Australian Rules football-hack activities and training resources through application.



About AFL

The AFL is Australia's premier sporting organization, supporting a constantly evolving national Australian Rules football competition. Its purpose is to progress the game, so everyone can share in its heritage and possibilities.

Industries: Media & Entertainment

Location: Australia

Google Cloud results

- Builds on Google's partnership with AFL to help the organization meet its objective of progressing the game so everyone can share in its heritage and possibilities
- Broadens accessibility to Australian Rules skills and training to users with visual and hearing needs
- Helps the AFL encourage the emerging talent that

© TensorFlow (<https://www.tensorflow.org/>)

Google structures the ML models that power the Footy Skills Lab to provide a great user experience

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Few sights in sport are as exhilarating as an elite Australian Rules footballer soaring above teammates and opponents to grab the football or launching an accurate 50 meter-plus kick for goal. These elements make Australian Rules football a wildly popular sport across most states and territories and internationally. According to a 2019 census conducted by the AFL (<https://www.afl.com.au/>), the peak body for Australian Rules football in Australia, approximately 530,000 people aged 13–39, or about 5.8% of the population in that age group based on the 2016 Australian Bureau of Statistics Census, participate in the game.

The game is also immensely popular among children, with 354,000 participants aged 5–12 (14% of the population in that age group based on the 2016 census) in Junior footy and the Auskick modified rules program.

Matches at the elite level last more than 90 minutes and are a searching test of players' skill, fitness, and ability to execute under pressure. As well as accurate kicking over long and short distances and spectacular "marks," players need to handpass by

holding the ball in one hand and punching it to a teammate, bounce an oval-shaped football while sprinting, tackle opponents effectively, and combine with teammates to break down opposition teams' defensive structures.

Teams that compete in the AFL's national competition, including the Melbourne Demons, Western Bulldogs, Richmond Tigers, Collingwood Magpies and Geelong Cats in Victoria, the West Coast Eagles in Western Australia, the Brisbane Lions in Queensland, the Sydney Swans in New South Wales, and the Adelaide Crows in South Australia are household names. Matches between well-supported rivals during the season may attract more than 80,000 spectators, while the Grand Final—held typically in September at the Melbourne Cricket Ground—has attracted more than 100,000 attendees on multiple occasions.

Working with Google to improve participation

The AFL defines its purpose as to “progress the game, so everyone can share in its heritage and possibilities.” This means addressing inequalities in access to elite Australian Rules football training techniques and skills development, coaching, and participation. The AFL opted to work with Google—a partner of three years standing—to improve access to coaching and community participation, regardless of ability, gender, location, culture, or socio-economic background.

A solution arrived in the form of work being undertaken by the Google Partner Innovation team, a group that works with partners and internal product teams to drive groundbreaking usage of Google Cloud (<https://cloud.google.com/>) technologies. The team had conducted work in cricket-mad India on user behaviors and became aware that children in rural areas were relying on YouTube videos to mimic the batting and bowling techniques of their favorite stars. This prompted team members to investigate how technology can improve sporting skills and evaluate how they could apply machine learning and the TensorFlow (<https://www.tensorflow.org/>) open source library to the opportunity. The team initially used PoseNet, a pose estimation model that estimates the spatial locations of key body joints—keypoints—on an image or video to monitor how subjects performed actions and identified opportunities to correct or improve these. In April 2019, fans of the Indian Premier League (IPL) cricket competition engaged with the YouTube Trainer installation during the IPL Trophy Tour to measure their batting skills against highlight footage of their favorite IPL players.

The Partner Innovation team further developed its athlete technique-related machine learning capabilities by working with the TensorFlow team in the United States to develop a mobile application called Dance Like That. This application uses a range of machine learning models to accurately compare users' movements with those of professional dancers and provide feedback accordingly. After successfully launching the application, the Partner Innovation team

collaborated with the Google Arts and Culture team on the [Australia: Great Sporting Land](https://artsandculture.google.com/project/australian-sports) (<https://artsandculture.google.com/project/australian-sports>)

launch in Melbourne in 2019 to create installations that featured skills demonstrations by AFLW (AFL Women's) star Tayla Harris and former Australian team cricket captain Steve Waugh.

Delivering content and coaching experiences over mobile applications

In November–December 2020, the team began working on a solution that would deliver content and coaching experiences to fans and players and opted for a scalable mobile application. The application had to work seamlessly on old phones as well as the latest models and over a range of networks, so users in remote areas such as Far North Queensland and northwestern Western Australia could receive the same experience as users in Melbourne or Sydney.

After experimenting with the PoseNet model, Google determined that tracking the movement of an Australian Rules football rather than a user would be most effective. The team reformed its approach to track the movement of a ball within a phone camera frame and now runs two TensorFlow-developed machine learning models to support the web application—one model trained with hundreds of AFL footballs of various colors in a range of lighting conditions to search the frame for a ball and one

model that, once a ball is detected, monitors the space and accounts for the position of the ball.

Using two machine learning models rather than a single model for ball detection and tracking reduces processing loads and time, ensuring a high-quality user experience on a range of devices.

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—Nicole Livingstone, AFL
General Manager of
Women's Football

The AFL and Google launch the Footy Skills Lab

On July 8, 2021, the AFL and Google launched the Footy Skills Lab (<https://play.afl/footyskillslab/>) to help budding footballers in Australia and worldwide sharpen their skills straight from their smartphone. The Footy Skills Lab is a free platform that helps players improve their skills through activities in ball-handling, decision-making, and kicking across three levels of difficulty. Tayla provides tips on kicking, while fellow AFLW stars Madison Prespakis and Akec Makur Chuot are among the AFLW and AFL Wheelchair athletes to deliver advice.

Audio prompts and closed captioning make tips and activities accessible for people with visual and hearing needs, while the Lab delivers a scorecard at the end of each activity that users may share with friends, family, teammates, and coaches.

AFL General Manager of Women's Football Nicole Livingstone says the launch of Google's Footy Skills Lab is another exciting way that the League continues to support emerging talent. "It is great to see our AFLW stars at the forefront of this technology. Having the Google Footy Skills Lab so accessible will help enhance the performance of the next generation of football talent," Ms Livingstone says.